

2207

جمهورية مصر العربية
وزارة التربية والتعليم والتعليم الفني
امتحان دبلوم مدراس التكنولوجيا التطبيقية الصناعية (نظام الثلاث سنوات)
امتحان - عام 2024

الزمن: ساعتان

المادة : Physics
التخصص : جميع التخصصات الصناعية

(2207)

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نموذج اجابة / عام 2024

التخصص : جميع التخصصات الصناعية (We & HST)

اسم المؤهل	دبلوم مدراس التكنولوجيا التطبيقية الصناعية (نظام الثلاث سنوات)
المادة	Physics
الوحدات التدريبية	From unit one till unit 4 (two terms each term contains 2 units)
المعارف والمهارات المكتسبة	Unit One (U1) : magnetic effect of electric current Unit Two (U2): electromagnetic induction Unit Three (U3): modern physics Unit Four (U4) : modern electronics

Group 1 :

A) Write the scientific term expressed by the following statements:

- 1) The magnetic flux density.
- 2) The magnetic dipole moment

B) Mention the factors affecting each of the following (one factor only)

- 3) The magnetic force acting on a wire carrying current placed perpendicular to a uniform magnetic field.

- a) The length of the wire (L) $F \propto L$ b) The current in the wire (I) $F \propto I$
c) The magnetic flux density (B) $F \propto B$

- 4) The density of magnetic field at a point along interior axis of a solenoid carrying an electric current.

- a) I: intensity of current passing through the coil b) L: the length of the solenoid
c) μ : Permeability of medium d) N: the number of turns of wire in the solenoid

C) Put circle around the correct answer

- 5) a) 10^{-2} Wb 6) c) $\frac{1}{1}$

D) Answer the following question

7)	An electric current of intensity 10 A is flowing in a long straight wire. Calculate the density of magnetic field at a distance 0.1m from the wire (Permeability of air = $4\pi \times 10^{-7}$ Wb /A .m)
8)	
$B = \frac{\mu I}{2 \pi d} = \frac{4 \pi \times 10^{-7} \times 10}{2 \pi \times 0.1} = 2 \times 10^{-5} T$	

E) Compare between each of the following

9 10	Point of comparison	Fleming left hand rule	Ampere's right hand rule
	The use	determine the direction of the force (motion) of a straight wire carrying current	determine the direction of the magnetic field resulting from an electric current in a wire or

Group 2 :

A) Write the scientific term expressed by the following statements:

11) Efficiency of a transformer

12) Electromagnetic Induction

B) Mention the factors affecting each of the following (one factor only)

13) The coefficient of mutual inductance between two coils

- a) The presence of an iron core inside the coil.
- b) The volume of the coil and the number of its turns.
- c) The distance separating them.

14) The instantaneous value of induced emf generated in the dynamo

- a) Area of the coil
- b) Number of turns
- c) Magnetic flux density
- d) Frequency (speed of rotation)
- e) The angle between normal on coil's plane (axis of the coil) and magnetic flux lines

C) Put circle around the correct answer

15) b) 30°

16) d) the resistance of magnetic path

17) The direction of the induced current will be such that it opposes the change in magnetic flux

D) Correct the underlined in the following statement

18) (Decrease or minimize)

19) (Electromagnetic induction)

20) (4 times)

Group 3:

A) Put sign (✓) in front of the correct statement or sign (X) in front of the incorrect statement of each of the following:

21) (✓)

22) (X)

23) (X)

C) Put circle around the correct answer

24) b) Thermionic emission

25) c) have the same speed

26) d) the intensity of the radiation increases

C) Mention the factors affecting each of the following (one factor only)

27) The kind of target element **oR** the atomic number of the target element

28) The temperature of the hot body

D) Write the mathematical law used to calculate

29) $P_L = \frac{E}{c} = \frac{h}{\lambda} = \frac{h\nu}{c} = mc$

30) K. $E_{\max} = h\nu - E_w = 0.5 m\nu^2$

Group 4:

A) Write the scientific term expressed by the following statements:

31) Thermal equilibrium (dynamic equilibrium)

32) Current gain

B) Put circle around the correct answer

33) c) pentavalent

34) a) increases

35) c) NAND gate

C) Give a reason for each of the following

36) Because all bonds in the crystal are unbroken) and has a high resistance to current flow (no free electrons at 0K)

37) To decrease the chance of recombination OR to decrease the portion lost in the base by recombination

D) Put sign (√) in front of the correct statement or sign (X) in front of the incorrect statement of each of the following:

38) (√)

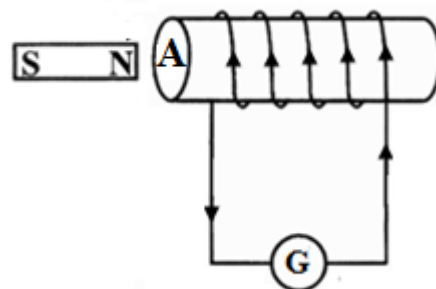
39) (X)

40) (X)

Group 5:

Answer the following

41)	A photon of wavelength 5.6×10^{-7} m falls on a metallic surface of work function
42)	3.42×10^{-19} J, calculate the kinetic energy of emitted electron from the metallic surface . (Planck's constant = 6.625×10^{-34} J.s, speed of light in air = 3×10^8 m/s) $K.E_{max} = h\nu - E_w$ $K.E_{max} = \frac{h c}{\lambda} - E_w = \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{5.6 \times 10^{-7}} - 3.42 \times 10^{-19}$ $K.E_{max} = 1.29 \times 10^{-20} \text{ J}$
43)	A metallic ring is placed in a magnetic field which changes from 0.008 Wb to
44)	0.004 Wb in 2s. Calculate the induced emf in the coil $\text{emf} = \frac{-N \Delta \Phi_m}{\Delta t} = \frac{-1 \times (0.004 - 0.008)}{2} = 2 \times 10^{-3} \text{ V}$
45)	Mention without explanation the two conditions necessary for laser action
46)	1) Population inversion 2) Stimulated emission
47)	Mention two factors only affect The magnetic torque acting on a coil
48)	1) magnetic flux density ($\tau \propto B$) (at constant other factors) 2) the current intensity ($\tau \propto I$) (at constant other factors) 3) the area of the coil's plane ($\tau \propto A$) (at constant other factors) 4) number of turns of the coil ($\tau \propto N$) (at constant other factors) 5) the angle between perpendicular to coil's plane and the magnetic flux lines
49)	In the opposite diagram a magnet is moving
50)	relative to the solenoid , look well at the opposite diagram then determine the following a) north pole (N) b) the magnet is moving towards the coil



(انتهت الاجابة)